

Periphrastic negation in Kalaallisut

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1 Introduction

Kalaallisut (ISO 639: kal, Inuit, Greenland) has two ways of forming negation: a negative suffix (1) and a periphrastic negation construction (2), which we refer to as the PNC.^{1,2,3,4}

- (1) Mersu-**nn**gi-langa.

sew-NEG-1SG:IND:NEG

‘I don’t sew.’

Ellen Thrane CRS-1

- (2) Mersor-**neq** ajor-punga.

sew-NEQ HAB:NEG-1SG:IND

‘I don’t sew.’

Ellen Thrane CSR-6

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²Elicited and volunteered examples are annotated with the name of the speaker and an alpha-numerical code identifying it in Line Mikkelsen’s notebooks. Textual examples primarily come from a collection of Kalaallisut texts transcribed and glossed by Maria Bittner (<https://sites.google.com/view/maria-bittner/kalaallisut>). These are annotated with the name of the story, paragraph number, and line number. Examples drawn from other online resources are sourced with a URL. Examples from Bittner’s text collection are rendered in standard orthography and reglossed in accordance with footnote 3.

³Glossing conventions: 1: first person; 2: second person; 3: third person; 4: fourth person (non-local person DPs that are co-referential with a higher subject are fourth person); ABL: ablative case; ABS: absolutive case; ALL: allative case; AP: antipassive; APPL: applicative; CAU: causative mood; CAUS: causativizer; COND: conditional mood; CONJ: conjunction; CONT: contemporative mood; COP: copula; DEF: definite; DEM: demonstrative; DIST: distal; EQUA: equative case; ERG: ergative case; FOC: focus; FRUST: frustrative; FUT: future; HAB: habitual; IND: indicative mood; INF: infinitive; INST: instrumental; INTRANS.PART: intransitive participle; LOC: locative case; NEG: negative; NOM: nominalizer; PART: participial mood; PASS: passive; PASS.PART: passive participle; PL: plural; PRES: present; PTCP: participle; SG: singular; VIA: vialis case. 3SG>3SG:IND means a third person singular subject acts on a third person singular object in indicative mood. 1SG>3SG:ERG means a first person singular possessor with a third person singular possessum in the ergative case. Suffixal negation triggers a special negative allomorph of the following mood-agreement morpheme. This is indicated :NEG in the glossing of the mood-agreement morpheme. As for *-neq*, Bittner (1994) glosses it GER for ‘gerund’, Fortescue (1984) glosses it NOM for ‘nominalizer’, and Kahn and Valijärvi (2022) gloss it AP for ‘abstract particle’. Since the category status of *-neq* is a central issue of the paper, we simply gloss it as NEQ and rely on the discussion in sections 4–6 to clarify its status. Note that clauses unmarked for tense can receive either a present or past interpretation, including in the PNC. Other abbreviations and notations used are: apP: Antipassive Phrase; ApplP: Applicative Phrase; C: complementizer; CausP: Causative Phrase; CP: Complementizer Phrase; D: Determiner; DP: Determiner Phrase; Neg⁰: Negation head; NegP: Negation Phrase; PNC: Periphrastic Negation Construction; PRO: null controlled subject DP; pro: null pronoun; SOV: Subject-Object-Verb order; Subj: Subject; V: Verb; VP: Verb Phrase; Voice⁰: Voice head (active or passive); VoiceP: Voice Phrase.

⁴A configuration of non-finite verb and finite negative auxiliary, as in (2), is also found in other languages, prominently Uralic languages, where it is known as a ‘connegative’ construction. See typological discussion of negative strategies at the end of this section.

Suffixal negation is the general purpose negation strategy, whereas the PNC expresses habitual negation (section 3.2). The suffixal strategy is unremarkable given the polysynthetic nature of the language (Fortescue 1999). The periphrastic construction is more surprising and our focus here. We develop an analysis in which *ajor-* is a negative auxiliary and the *-neq* suffix a non-finite verbal element. Evidence for this analysis comes from a systematic comparison with control structures of the type illustrated in (3), which also feature a *neq*-suffix verb followed by a finite verb.

- (3) Naja-p mersor-**neq** nuannari-vaa.
 Naja-ERG sew-NEQ love-3SG>3SG:IND
 ‘Naja loves to sew.’

Grethe Schmidt CIC-GS-1

While both constructions employ the *-neq* suffix, we argue that it instantiates different syntactic categories in the two structures. In the PNC, *-neq* is a non-finite T head. In control structures, it is a D head, in line with its general nominalizing function (Fortescue 1984:44ff). We present morphological, syntactic, and semantic evidence for this distinction from case marking, verbal agreement, possessive inflection, scrambling, subcategorization, and theta roles. While the PNC is documented in descriptive and pedagogical work (Fortescue 1984:47–48, Kahn and Valijärvi 2022:245–247, Sadock 2003:10–11,33, Woodbury 1975:70–71), the present paper provides, to the best of our knowledge, the first in-depth analysis of the PNC and the first systematic comparison with control structures.

In support of our synchronic analysis, we provide a diachronic analysis of the development of the PNC that situates it in a larger group of ‘negative verbs’. We argue that *ajor-* originated as a transitive control verb meaning ‘be unable to reach or do’ that took a nominalized DP complement, akin to *nuannari-* ‘love’ in (3). Two intertwined changes then occurred: the nominalizer *-neq* was reanalyzed as a verbal morpheme and *ajor-* was grammaticalized into a negative auxiliary verb. We present evidence for this analysis from Proto-Inuit, from comparison with other modern day Inuit varieties, and from observations internal to the grammatical system of Kalaallisut.

In addition, our demonstration that *ajor-* is a negative auxiliary is of typological relevance. Negative auxiliaries form a rare negation strategy. According to the classification in WALS (Feature 112A), only 47 of 1157 languages, or 4%, use negative auxiliaries. Examination of the original sources for these 47 languages reveals that five of these languages are like Kalaallisut in also having a negative affix in addition to a negative auxiliary (see the appendix for details). WALS additionally lists 21 languages with ‘variation between a negative word and affix’. Examination of these 21 languages reveals that in one of them, Lavukaleve (Papuan, Solomon Islands), the negative word is an auxiliary. There thus appears to be only six languages in the sample (.5%) with the same negation profile as Kalaallisut. Of these six only three have the same morphological profile (polysynthetic) as Kalaallisut (details in the appendix).⁵ Thus, the Kalaallisut negation system is quite rare and hence of typological interest.⁶

In section 2, we provide background on our collaboration and on Kalaallisut language ecology as well as relevant linguistic features of the language. Section 3 introduces the PNC by illustrating

⁵Thanks to Arly Madeline Siu for checking original sources for the 47 languages listed in WALS as having a negative auxiliary and to Jamison Allen Damiano for identifying Lavukaleve as the only language with a negative auxiliary in 21 languages listed WALS as having a negative word in addition to a negative affix.

⁶WALS classifies Kalaallisut as a language with only a negative affix so our documentation of a second negation strategy, a negative auxiliary, suggests a reclassification of the language into the ‘variation between negative word and affix’ class.

its function in language use in a variety of genres and by comparing it to suffixal negation. We also discuss the homophonous lexical verb *ajor-* ‘be bad’ and argue that it is distinct from the negative auxiliary. Section 4 lays out the empirical differences between the PNC and control structures that motivate an analytic distinction between the two. In section 5, we examine a number of predictions this analytic distinction makes and show that they are borne out. This then paves the way for our structural analysis of the two constructions in section 6. Here we locate the *-neq* suffix on the clausal spine and present arguments for *ajor-* being a negative auxiliary located in Neg. In section 7, we turn to diachrony. Section 8 concludes the paper.

2 Background

This section provides background on the collaboration behind this paper (section 2.1), lays out the linguistic ecology of Kalaallisut (section 2.2), and exemplifies basic linguistic features of Kalaallisut that are relevant for the analysis of the PNC and control constructions at the heart of the paper (section 2.3).

2.1 Collaboration

This work stems from a collaboration between linguists Line Mikkelsen and Tyler Lee-Wynant and Kalaallisut speakers Grethe Schmidt and Ellen Thrane.

Grethe Schmidt’s autobiography: I was born in Attu in northwestern Greenland and spoke Kalaallisut at home and in school. I lived in a number of other places in Greenland before moving to Denmark in 1980. I am a Kalaallisut-Danish-Kalaallisut translator for Danish TV channels and also affiliated with Umiaq, a Greenlandic association hosted at Kalaallit Illuutaat (‘The Greenlanders’ house’) in Odense, Denmark. Kalaallisut is the language I am born with.

Ellen Thrane’s autobiography: I was born in Upernavik in northwestern Greenland. My mother and grandmother spoke only Kalaallisut, whereas my father spoke Kalaallisut and Danish. I acquired Kalaallisut from birth and only started learning Danish in school. I moved south to Ilulissat at age 9 and to Denmark after 7th grade for educational opportunities. I volunteer as a Kalaallisut-Danish-Kalaallisut interpreter for Danish public agencies and I am affiliated with Umiaq at Kalaallit Illuutaat. Kalaallisut is my mother tongue and identity.

Line Mikkelsen and Ellen Thrane began collaborating in summer of 2022, Grethe Schmidt joined the project in spring of 2024, and Tyler Lee-Wynant in Fall of 2024. The data discussed below come from one-on-one weekly Zoom elicitation sessions, which also included discussion of linguistic questions of interest. Elicitation was conducted by Line Mikkelsen in Danish. Danish prompts and translations were subsequently translated into English by Line Mikkelsen. All elicitation recordings are archived in the California Language Archive. With the exception of the scrambling data in (44), judgments were overall uniform within and between speakers. Key data points were additionally checked with Kalaallisut speaker Agnes Johnsen and native-speaker linguist Naja Blytmann Trondhjem.

2.2 Language ecology

Kalaallisut is a member of the Inuit–Yupik–Unangan language family, which spans a region from Eastern Russia, across Alaska and Canada, to Greenland. It is the sole official language of Greenland, with around 50,000 speakers in Greenland and another 10,000 in Denmark.⁷ The language is acquired by children in the home and taught in school. Denmark has been and continues to be a colonial power in Greenland; the Danish language (ISO 639: dan) has a big presence, especially in administration. English (ISO 639: eng) has a growing presence.

2.3 Basic linguistic features

Three linguistic aspects of Kalaallisut are of immediate relevance for our study: i) polysynthesis, ii) case, and iii) verbal agreement.

Kalaallisut manifests classic features of polysynthesis. First, verbs may exhibit a high morpheme-to-word ratio, as in the textual examples in (4)–(6).⁸

- (4) Qulingilua–a–su–ssa–a–galuar–put.
nine–COP–NOM–FUT–COP–FRUST–3PL:IND
‘There should’ve been nine.’ *Paakujuk* (Sooty), §10, (11)
- (5) Nukappiaqqa–p unit–sin–niar–aluar–pai.
boy–ERG stop–CAUS–try–FRUST–3SG>3PL:IND
‘The boy tried to stop them [= the dogs].’ *Paakujuk* (Sooty), §27, (2)
- (6) ... navia–nar–tor–sior–tip–paanga.
fear–induce–NOM–experience–CAUS–3SG>1SG:IND
‘... it put me in danger’ (lit. ‘it made me experience having fear induced’)
Piniartup uqaluttuaa (A hunter’s story), §5, (4)

As these examples also show, there is rich verbal inflectional morphology, specifically mood and subject and object agreement. These categories are expounded together in a single, obligatory word-final suffix. Third, Kalaallisut exhibits grammatically free argument drop and noun incorporation.⁹ This is illustrated in (7), where all DP arguments are null pronouns and the notional object of ‘consume’ is incorporated.

- (7) *pro* kaagi–sor–uit *pro pro* ajori–ssu–akkit.
2SG:ABS cake–consume–2SG:COND 1SG:ERG 2SG:ABS angry–FUT–1SG>2SG:IND
‘If you eat cake, I’ll be angry with you.’ Ellen Thrane ANM-1

Finally, Kalaallisut basic word order is SOV, but with discourse-induced flexibility (Fortescue 1984:93ff; Fortescue 1993).

⁷The number of Kalaallisut speakers in Denmark reported above comes from (<https://greenlandic.dk/om-sproget/>) and is based on reports from the “Greenlandic Houses” and linguistic studies by Ilisimatusarfik (University of Greenland). There is no official government tally of Kalaallisut speakers in Denmark.

⁸A reviewer points out that the word for ‘nine’ is, at least historically, morphologically complex, deriving the literal meaning of ‘not quite ten’. Taking this into account would increase the morpheme-to-word ratio in (4) even further.

⁹A distinctive feature of Inuit noun incorporation is that it is lexically restricted to a set of obligatorily incorporating verbal roots.

Case alignment is ergative-absolutive. In (8) the transitive subject is marked with the ergative suffix *-p*. In contrast, the transitive object and intransitive subject (9) are absolutive, which is null. The absolutive/ergative case distinction is only morphologically expressed in singular and possessed nouns. We nonetheless gloss all core arguments as ergative or absolutive in accordance with their syntactic function.

- (8) Naja-**p** Juuna ikior-paa.
 Naja-ERG Juuna:ABS help-3SG>3SG:IND
 ‘Naja helped Juuna.’ Ellen Thrane AFX-1

- (9) Juuna angerla-jaar-poq.
 Juuna:ABS come.home-early-3SG:IND
 ‘Juuna came home early.’ Ellen Thrane AFX-2

There is at most one absolutive argument per clause (Compton 2018:164). Therefore, two internal arguments of a ditransitive verb cannot both be absolutive (10). Instead, one must be realized as an oblique (11).

- (10) *Meeqqa-t poortukka-t nassip-pakka.
 child-PL:ABS package-PL:ABS send-1SG>3PL:IND
 Intended: ‘I sent the children the packages.’ Ellen Thrane CKS-2

- (11) Meeqqa-t poortukka-nik nassip-pakka.
 child-PL:ABS package-INST:PL send-1SG>3PL:IND
 ‘I send the children the package.’ Ellen Thrane CKS-3

Verbs agree with all absolutive and ergative arguments. (12) is ungrammatical because the verb only agrees with the subject. (13) is ungrammatical because the verb only shows agreement with the object. The only grammatical realization of a formally transitive clause is as in (14) where the verb agrees with both the ergative subject and the absolutive object.

- (12) *Naja-p meeqqa-t ikior-poq
 Naja-ERG child-PL:ABS help-3SG:IND
 Intended: ‘Naja helped the children.’ Grethe Schmidt CIE-GS-1

- (13) *Naja-p meeqqa-t ikior-put.
 Naja-ERG child-PL:ABS help-3PL:IND
 Intended: ‘Naja helped the children.’ Grethe Schmidt CIE-GS-2

- (14) Naja-p meeqqa-t ikior-pai.
 Naja-ERG child-PL:ABS help-3SG>3PL:IND
 ‘Naja helped the children.’ Grethe Schmidt CIE-GS-3

Verbs do not agree with oblique arguments, including allative (15) and instrumental (16) arguments:

- (15) Ilin-nut allap-punga/*pakkit.
 2SG-ALL write-1SG:IND/1SG>2SG:IND
 ‘I wrote to you.’ Ellen Thrane CHT-1/CHT-2

- (16) Meeqqa-nik iku-i-vunga/*vakka.
 child-INST.PL help-AP-1SG:IND/1SG>3PL:IND

Intended: ‘I helped some children.’

Grethe Schmidt CZJ-GS-3/CZJ-GS-2

In sum, verbs agree with all and only absolutive and ergative arguments.

3 Introducing the Periphrastic Negation Construction

In this section we introduce the PNC in a general way as background for the more targeted description and analysis in the following sections. We start by illustrating the use of the PNC (section 3.1). We then compare it to affixal negation in terms of semantics (section 3.2) and frequency (section 3.3). Finally, we discuss the relationship between the negative auxiliary *ajor-* and the homophonous lexical verb meaning ‘be bad’ (section 3.4).

3.1 PNC usage

While the PNC is less frequent than suffixal negation (see section 3.3 below), it is robustly attested across all genres that we have access to.¹⁰

The example in (17) comes from *Silliarnamik uqaluttuaq* (The tale of Silliarnaq) as included in Maria Bittner’s online text collection (<https://sites.google.com/view/maria-bittner/kalaallisut>). Bittner characterizes this text as a myth. The myth follows three brothers and their friend. On a hunting trip they end up far out on the ice-covered sea. They eventually arrive in an unfamiliar land, where an old man becomes their foster father. In the translation of the sentence below, ‘they’ refer to the brothers and their friend and ‘the old man’ to the foster father. The habitual meaning of the periphrastic negation is consonant with the context, specifically the use of the habitual suffix *-(g)aang* in the preceding subordinate clause (*whenever they woke up in the morning*.)

- (17) a. Ukiu-p ingerla-ner-a-ni ullaa-kkut iter-**aang**-ata
 year-ERG go.ON-NEG-3SG>3SG-LOC morning-VIA wake.up-HAB-3PL:CAU
 ‘In the course of the year, whenever they woke up in the morning,’
 b. utoqqaar-suaq **peqar-neq ajor-poq**
 old-big:SG:ABS be.there-NEG HAB:NEG-3SG:IND
 ‘the old man was never there.’

Silliarnamik uqaluttuaq (The tale of Silliarnaq), §9, (1)

The example in (18) is from a personal narrative by Ellen Thrane. She is narrating about her childhood, including play and other activities. As in (17) the habitual component of the PNC is matched by a habitual morpheme, *-tar*, located in the following contrastive clause.

¹⁰The English translations of (17) and (21) are from Bittner’s text collection. Examples (18)–(20) and (22)–(24) were translated into Danish by Grethe Schmidt and Ellen Thrane and then into English by Line Mikkelsen.

- (18) Taamaammat **sisora-neq ajor-pugut** arpa-**tar**-pugut.
 therefore ski-NEQ HAB:NEG-1PL:IND run-HAB-1PL:IND
 ‘Therefore we don’t ski as much, but run instead.’

Ellen Thrane, personal narrative, March 2024.

The example in (19) comes from a social media post. It is clear from the context that the author is conveying that they generally don’t have time for dating, not that there is a particular date that they do not have time for.

- (19) Aappa-qar-ni-ssa-mut **piffissa-qar-neq ajor-punga** ...
 mate-have-NOM-ALL time-have-NEQ HAB:NEG-1SG:IND
 ‘I don’t have time for dating ...’

Aleqa Hammond (Facebook)

The PNC is also found in children’s books as illustrated by the example in (20) taken from *Ataata isertitsivimmiippoq* (Father is in jail.). Here the PNC negates Sofus’s expectation that his father would be regularly sending gold back to his family.

- (20) Sofus lives with just his younger brother, older sister and mother. ... Sofus_i misses his_i father_k because he_k has gone to Naluna to dig for gold.

Sofusi-l=li paasi-sinnaa-nngi-laa sooq guldi-nik **nassitsin-neq**
 Sofus-ERG=but understand-can-NEG-3SG>3SG:IND:NEG why gold-INST.PL receive-NEQ
ajor-lutik ataata-miniit.
 HAB:NEG-4SG>4PL:CONT father-4SG>3SG:ABL

‘But Sofus can’t understand why they [Sofus, his siblings and mother] don’t receive gold from his father.’

<https://milik.gl/ka/milik/e-bogit/>

The next example comes from a description of the bird species *appaliarsuk* (Little Auk; Alle alle) and its way of life. It is drawn from a passage explaining how it dives for krill. The habitual semantics of the PNC in (21) is conveyed by the use of ‘never’ in the English translation.

- (21) Qut-sis-su-kkut **ingerla-neq ajor-put.**
 top-be.at-3SG:PART-EQUA travel-NEQ HAB:NEG-3PL:IND
 ‘They [the Little Auks] never fly high
 ... while feeding, first when they’ve finished feeding do they fly up and go home to roost.’

Appaliarsuit (Little auks), §2, (2)

Example (22) comes from the public broadcasting company KNC (Kalaallit Nunaata Radioa; lit. Greenlanders’ country’s radio) and occurs in a description of a town, whose inhabitants are committed to keeping the town clean. Here the habitual semantics of the PNC is reflected in the English simple present tense translation.

- (22) Maani inuusuttut utoqqaa-l=lu maangaannaq igitassa-nik
 here young.person-PL:ABS old-person-PL:ABS=and for.no.reason garbage-INST:PL
igitsi-neq ajor-put, ...
 throw.away-NEQ HAB:NEG-3PL:IND
 ‘Here young and old people don’t throw garbage for no reason ...’

<https://kni.gl/nunaqarfiga-kusanartog/>

Finally, the PNC is amply attested in newspapers (23) and other non-fiction writing (24).¹¹

- (23) Amerlaqi-su-tigut **ilinniaga-qar-neq ajor-put,** ...
 be.many-NOM-VIA:PL education-have-NEQ HAB:NEG-3PL:IND
 ‘Many [children growing up in poverty] don’t get an education, ...’
<https://tinyurl.com/2z6scc4n>
- (24) Sisussat¹² sumiorpaluute-qar-palu-mmat taanna **ator-neq ajor-paat.**
 Sisussat dialect-have-seems-3SG:CAU DEM:DIST use-NEQ HAB:NEG-3PL>3SG:IND
 ‘Because Sisussat seems to have a dialect, they don’t use that [term].’
<https://tinyurl.com/bdt7xt6t>

These examples show that the PNC is found in both formal and informal language use and in both traditional language use and current language use. One thing missing from this list of examples above is PNC use in conversation. We currently do not have access to a conversational corpus and also have not had the opportunity to listen for the PNC in conversation. That said, there are reasons to think that the PNC is also used in conversation. First, is it frequent in social media posts, e.g. (19), which is characterized by informality, as is casual conversation. Second, the PNC is sometimes produced in Danish-to-Kalaallisut oral translation tasks in elicitation and also in spontaneous oral narratives, as in (18) above.

3.2 Semantics

As seen in the examples above, the PNC generally expresses habitual negation. This is in contrast with the negative suffix *-nngit*, which expresses ‘plain’ negation and is used in a wide range of semantic contexts, illustrated in the examples below.

First, the suffixal negation example in (25) forms a minimal pair with the PNC example in (17). In (17), the old man is habitually absent (every morning). By contrast, (25) occurs in the telling of a particular episode of the myth *Aataarsuup irnikasia* (Aataarsuaq’s kid son), where Aataarsuaq is in pursuit of his son. Aataarsuaq takes his eyes off his son for just a moment and when he looks again his son is gone.

- (25) Taamak peqa-**nngi**-laq.
 gone be.there-NEG-3SG:IND:NEG
 ‘Gone, he [Aataarsuaq’s kid son] wasn’t there.’
Aataarsuup irnikasia (Aataarsuaq’s kid son), Part IV, §1, (9)

¹¹In (23) we see the PNC in a generalizing statement. We are not quite sure how to characterize this in terms of habitual semantics, but we include it here to show some of the nuances of the meaning of the PNC.

¹²This is a local term for a place where a dog sled slides down a steep incline. It also refers to the inhabitants of that place.

Other representative examples of suffixal negation are included in (26)–(30) below. Collectively these suggest that *-nngit* is a general purpose negator. It can negate various event types, including change of state events (26), semelfactives (29), actions ((27) and first clause of (28)) and states (second clause of (28), (30)). It is compatible with modals (27) and can occur in all eight moods, here illustrated with indicative (27), contemporative (26), and conditional (28) moods.

- (26) Suli Tiggaa–p paata–i ersig–unnaa–**nngi**–llat,
still alpha.seal–ERG paddle.PL–3SG>3PL.ABS be.visible–cease–NEG–3PL:CONT.NEG
'As yet, Alpha's paddle hasn't become invisible,'
Mitinnguaq (Little Eider) §5, (3)
- (27) "Nalus–sinnaa–**nngi**–langa." aapakaaq oqar–poq.
swim–be.able–NEG–1SG:IND:NEG monkey say–3SG:IND
'"I can't swim.", said the monkey.'
Aapakaaq kuukkuuriarlu (The monkey and the crocodile) §4, (5)
- (28) Uku ila–kka ila–a–**nngik**–kaluar–unik suquta–a–**nngi**–laq.
these part–1SG>3PL:ABS part–be–NEG–FRUS–3PL:COND issue–COP–NEG–3SG:IND:NEG
'If these other pups don't come along, it doesn't matter.'
Paakujuk (Sooty) §6, (5)
- (29) Paa–kujuk qilu–ria–qqi–**nngi**–laq.
soot–be.like bark–stage.one–again–NEG–3SG:IND:NEG
'Sooty didn't give another woof.'
Paakujuk (Sooty) §15, (1)
- (30) Toqut–si–so–qar–**ta**–**nngi**–laq.
kill–AP–NOM–have–HAB–NEG–3SG:IND:NEG
'There are usually no murders.'

<https://oqaasileriffik.gl/da/sprogteknologi/korpus>

Example (30) is of special interest because it demonstrates that suffixal negation can co-occur with the habitual suffix *-tar*, making an alternative strategy for expressing habitual negation available. We have not investigated possible semantic differences between *ta-nngit* (HAB-NEG) and *ajor-*. Our point here is simply that suffixal negation is compatible with habitual semantics, but does not express habituality on its own.

3.3 Status of the PNC

A reviewer asks whether the PNC is marginal in the language. In a qualitative sense, the broad attestation of the PNC documented above suggests that it is not marginal. In terms of frequency, textual counts reveal that the PNC is significantly less frequent than suffixal negation. In the 14 texts curated by Maria Bittner (<https://sites.google.com/view/maria-bittner/kalaallisut>), there are 163 examples of suffixal negation and only 10 examples of the PNC. We don't think this entails that the PNC is marginal in the sense that it belongs at the edge of the grammatical structure of the language. Instead, we would like to suggest that the lower frequency arises from two factors. First, we hypothesize that communicative need to express habitual negation arises less often than

the need to express plain negation. If so, it stands to reason that the PNC is less frequent than suffixal negation. The second factor is that there is a morphological alternative to the PNC. As (30) above shows, habitual negation can also be expressed purely morphologically by combining affixal negation with the habitual morpheme *-tar*.

A third way that the PNC could be considered marginal would be if it were a calque from Danish. There are at least two reasons to think the PNC is not a calque. First, Danish has two ways of expressing habitual negation and neither resemble the PNC. The first strategy is to use the particle used for ‘plain’ negation (*ikke* ‘not’) together with a verb in the simple present tense. Danish

- (31) Hun drikke-**r** **ikke** the om morgen-en.
 she drink-PRES NEG tea about morning-DEF
 ‘She doesn’t drink tea in the morning.’ Line Mikkelsen

There is no mapping from this Danish construction to the PNC. In the Danish construction, negation is expressed by a particle, not a verb, and the lexical verb is finite, whereas it is non-finite in the PNC. The second strategy employs the raising verb *pleje* ‘usually do’ in combination with the negative particle.

Danish

- (32) Hun **pleje-r** **ikke** at drikke the om morgen-en.
 she usually.do-PRES NEG to drink-INF tea about morning-DEF
 ‘She usually doesn’t drink tea in the morning.’ Line Mikkelsen

This construction is a little closer to the PNC in that the lexical verb is non-finite. However, unlike the PNC, negation is expressed as an uninflected particle and the non-lexical verb expresses just habituality, as seen by the fact that it is also used in positive habitual statements (33).

Danish

- (33) Hun **pleje-r** at drikke the om morgen-en.
 she usually.do INF drink tea about morning-DEF
 ‘She usually drinks tea in the morning.’ Line Mikkelsen.

We thus see no way that either of the Danish constructions could have given rise to the PNC in language contact. Second, the PNC is found in myths, e.g. (17), whose composition presumably predates contact with Danish.

We conclude from the data and discussion in this section that the PNC is relatively infrequent, but not marginal in the grammatical system of the language.

3.4 Homophony of *ajor-*

As noted at the beginning of the paper, the form *ajor-* is also used as an intransitive verb with the meaning ‘be bad’. An example of this use is provided in (34).

- (34) kami-kasi-ni=lu **ajor**-luinna-ler-mata.
 kamik-wretched-4SG>3PL:ABS=and be.bad-absolutely-begin-3PL:CAU
 ‘and because his wretched kamiks had got really bad.’
Qillarsuakkunik oqalualaaq (Qillarsuaq’s Saga) §13, (1)

While space prevents us from providing a full discussion, we note that the sentence type in (34) is different from the PNC in at least four ways. First, there is no *-neq* verb form involved, just the nominal subject argument ‘his wretched kamiks’. Second, the subject uniformly receives its theta role from *ajor-*. In the PNC, by contrast, the theta role varies with the *neq*-marked lexical verb.¹³ Third, lexical *ajor-* is formally strictly intransitive—(34) vs. (35)—whereas the PNC exhibits transitive agreement when the *neq*-marked verb is transitive, e.g. (24).

- (35) *Siallu–p Juuna ajor–paa.
rain–ERG Juuna:ABS be.bad–3SG>3SG:IND
Intended: ‘The rain is bad for Juuna.’

Grethe Schmidt CZK-GS-2

The last thing to note is the difference in semantics: in the PNC *ajor-* expresses sentence negation and habituality, but not evaluation. In (34) *ajor-* expresses evaluation, but not sentence negation or habituality. The negative auxiliary *ajor-* is clearly synchronically distinct from the lexical intransitive verb *ajor-*. In terms of diachrony we argue in section 7 that auxiliary *ajor-* developed from a transitive verb *ajor-* ‘be unable to reach or do’, which has since fallen out of use. We have not investigated whether intransitive *ajor-* developed from the same transitive negative verb.

4 The duality of *-neq*

In addition to the PNC (36) *neq*-marked verbs feature in control constructions (37).¹⁴

- (36) Naja **biiler–neq** ajor–poq.
Naja:ABS drive–NEQ HAB:NEG–3SG:IND
‘Naja doesn’t drive.’
(37) Naja–p **biiler–neq** nuannari–vaa.
Naja–ERG drive–NEQ love–3SG>3SG:IND
‘Naja loves to drive.’

Ellen Thrane BPA-3

Ellen Thrane CHB-4

Despite their surface similarity, we argue that the status of the *neq*-phrase differs.¹⁵ In (36)

¹³Here we note in response to a reviewer that the PNC imposes no animacy restrictions on the subject. It is felicitous with non-human (21) and inanimate (i) subjects.

(i) Ujaqqa–t nakkakat–tar–neq ajor–put.
stone–PL:ABS roll.down–HAB–GER HAB:NEG–3PL:IND

‘The stones never roll down.’

Grethe Schmidt CUL-GS-3

¹⁴A reviewer asks whether control is necessarily involved in (37) or whether it could ‘simply be a nominalization’. This alternative analysis would work equally well with our account, because it preserves the key distinction between (36) and (37), namely that the *neq*-phrase is verbal in the former but nominal in the latter. That said, we do believe that there is evidence that control is involved in (37). Specifically, the truth of (37) hinges on Naja’s feeling about *herself* driving and cannot be followed by ‘... but Naja has never driven.’ This fact falls out from the control analysis, but not from a non-control nominalization analysis, where the agent of driving is not controlled by the matrix subject. For recent discussion of control into nominalizations see ?.

¹⁵The term *neq*-phrase refers to the constituent containing the *-neq* suffix, the non-finite verb, and any dependents of the non-finite verb.

biiler-neq is a non-finite verbal projection. In (37) it is a nominalization.¹⁶ There are three pieces of evidence that the *neq*-phrase is nominal in control structures but not in the PNC.

First, in control structures, the *neq*-phrase can bear possessive morphology (38); in the PNC it cannot, whether or not *ajor-* bears intransitive (39) or transitive (40) agreement.

- (38) Naja-p mersor-ni-**ni** nuannari-vaa.
 Naja-ERG sew-NEQ-4SG>3SG:ABS love-3SG>3SG:IND
 ‘Naja_i loves her_i sewing.’ Ellen Thrane CHR-1
- (39) *Naja mersor-ni-**ni** ajor-poq.
 Naja:ABS sew-NEQ-4SG>3SG:ABS HAB:NEG-3SG:IND
 Intended: ‘Naja doesn’t sew.’ Ellen Thrane CHR-2
- (40) *Naja-p mersor-ni-**ni** ajor-paa.
 Naja-ERG sew-NEQ-4SG>3SG:ABS HAB:NEG-3SG>3SG:IND
 Intended: ‘Naja doesn’t sew.’ Ellen Thrane CII-7

Second, control verbs like *nuannari-* ‘love’ can take underived nominal complements (41), the negative auxiliary *ajor-* cannot (42):

- (41) Sialuk nuannari-vara.
 rain:ABS love-1SG>3SG:IND
 ‘I love rain.’ Grethe Schmidt CKJ-GS-2
- (42) *Sialuk ajor-poq.
 rain:ABS HAB:NEG-3SG:IND
 Intended: ‘It doesn’t rain.’ Grethe Schmidt CKJ-GS-1

These differences in possessive morphology and subcategorization fall into place if *V-neq* is verbal in the PNC but nominal in control structures.

A third potential contrast between control constructions and the PNC comes from the claim in Sadock (2003:10,33) that in the PNC, the *neq*-marked verb must be immediately adjacent to the negative auxiliary. This is clearly not the case for control verbs. Examples like (43), where the *neq*-phrase has scrambled over the subject, are always judged grammatical.

- (43) Mersor-neq Naja-p nuannari-vaa.
 sew-NEQ Naja-ERG love-3SG>3SG:IND
 ‘Naja loves to sew.’ Grethe Schmidt CIC-GS-2

This word order difference would follow from the *neq*-phrase being a DP in control structures but a TP in the PNC, since DPs can scramble but TPs cannot (Grewendorf and Sternefeld 1990:13–14; Webelhuth 1992:180). However, in our work, scrambling of the *neq*-phrase in a PNC receives mixed judgments.

¹⁶This duality of *V-neq* is reminiscent of the duality of English *-ing* forms, which also have verbal and nominal guises (Abney 1987:167–244). The PNC most closely resembles the participle use of *-ing* as in *Naja is singing silly songs*. The Kalaallisut control construction in (37), however, has no correspondent in English. In Abney’s terms it is a blend of *PRO-ing* (*Naja_i likes PRO_i singing silly songs*) and *poss-of-ing* (*Naja likes Harvey’s singing of silly songs*). That blend is ungrammatical in English: **Naja_i likes PRO_i singing of songs*, suggesting that the analogy between Kalaallisut *-neq* and English *-ing* is not exact. While we agree that this contrast between nominal *-ing/-neq* constructions is worthy of further exploration, we will not attempt it here.

- (44) %Mersor-neq Naja ajor-poq.
 sew-NEQ Naja:ABS HAB:NEG-3SG:IND
 Intended: ‘Naja doesn’t sew.’

Grethe Schmidt rejected examples like (44) on four out of six occasions and Ellen Thrane rejected it on three out of five occasions. One possible interpretation of this state of affairs is that the ban on scrambling of TP, which was established mostly based on languages with a very different typological profile, is weaker in Kalaallisut and manifests as TP scrambling being of variable acceptability. Under that interpretation, the relative acceptability contrast between (43) and (44) supports our claim that in control structures the *neq*-phrase is a DP, whereas it is a TP in the PNC.

We thus propose that in the PNC *-neq* heads a non-finite verbal projection, specifically non-finite TP. This straightforwardly explains the verbal behavior of the *neq*-phrase in the PNC: it does not host possessive morphology, it does not alternate with DP complements, and scrambling is of variable acceptability. In control structures, *-neq* has its usual nominalizing function, which we take to be the manifestation of a D head.¹⁷ Being a DP, the *neq*-phrase in a control construction can host nominal morphology, alternates with a non-derived DP complement, and scrambles.

Before we turn to a more detailed examination of case and agreement properties of PNC and control structures, we briefly consider past treatments of the *neq*-suffix. All descriptions that we are aware of treat *-neq* as a nominalizing suffix, including Bittner (1994:53ff), Fortescue (1984:44ff), Nielsen (2021:199), and Olsen and Hertling (2011:48). More specifically Bittner (1994:53-54, 64) analyzes *-neq* as a noun in her account of control into adjunct clauses formed with *-neq* (see (55)). While this analysis can be extended to account for the nominal properties of *neq*-phrases in the complement control structures we are concerned with here, it doesn’t extend to *neq*-phrases in the PNC, which have none of these nominal characteristics. In sum, while some past descriptions of *-neq* have noted the PNC and its resemblance to complement control structures, none of them have provided an analysis that recognizes their differences when it comes to the function of *-neq*.

5 Case and agreement

The discussion above lets us posit the following schematic structures for the PNC (45) and control (46):¹⁸

- | | |
|--|---------|
| (45) Subj [_{TP} VP- <i>neq</i>] Aux | PNC |
| (46) Subj [_{DP} VP- <i>neq</i>] V | CONTROL |

Now we examine the predictions about case and agreement that follow from these structures together with general principles of case and agreement in the language. We start with control.

¹⁷Another analytic possibility is that *-neq* is a n head in control constructions (see Alexiadou (2020) for an overview of nominalization by D and by n). We will not attempt to argue n vs. D nominalization here, as the only thing that matters for us is that there is nominalization in control structures but not in the PNC. See Carrier and Compton (2024) for relevant discussion of the category of the cognate Inuktitut nominalizer *-niq*.

¹⁸In (45) and (46) VP stands in for the argument structure projections vP, Pass(ive)P, Caus(ative)P, Appl(icative)P, and a(nty)p(assive)P; see section 6.

5.1 Control

Our starting point for control structures is that the *neq*-phrase complement of the control verb is a DP. This makes five predictions about case and agreement which we consider in turn.

Prediction 1 As an internal DP argument of a transitive verb, the *neq*-phrase is absolutive. This claim is hard to assess in basic control constructions because absolutive case is null. However, possessive *neq*-DPs show distinctive absolutive morphology shared with possessed underived nominals (47) and (48). This absolutive possessive morphology contrasts with possessive morphology in other cases, including ergative possessive morphology in (49).¹⁹

- (47) Hansi-p qimmi-**ni** nuannari-vaa.
 Hansi-ERG dog-4SG>3SG:ABS love-3SG>3SG:IND
 ‘Hansi_i loves his_i dog.’ Grethe Schmidt CIH-GS-1
- (48) Naja-p mensor-ni-**ni** nuannari-vaa.
 Naja-ERG sew-NEQ-4SG>3SG:ABS love-3SG>3SG:IND
 ‘Naja loves to sew./Naja_i loves her_i sewing’ Ellen Thrane CHR-1
- (49) Hansi-p qimmi-**ata** kii-vaanga.
 Hansi-ERG dog-4SG>3SG:ERG bite-3SG>1SG:IND
 ‘Hansi’s dog bit me.’ Grethe Schmidt CIH-GS-2

Prediction 2 As the subject of a verb with an absolutive object, the subject is ergative (Yuan 2018). This holds true even if the embedded verb is intransitive, as seen in (50).

- (50) **Naja-p/*Naja** biiler-neq nuannari-vaa.
 Naja-ERG/Naja:ABS drive-NEQ love-3SG>3SG:IND
 ‘Naja loves to drive.’ Ellen Thrane CHB-4/CHB-5

Prediction 3 Since there is at most one absolutive DP per CP (Compton 2018:164), it should not be possible to realize an internal argument of the *neq*-marked verb as an absolutive. This is indeed impossible:

- (51) *Meeqqa-nut poortukka-t nassi-un-neq nuannari-vara.
 child-PL:ALL package-PL:ABS send-APPL-NEQ love-1SG>3SG:IND
 Intended: ‘I love to send packages to the children.’ Ellen Thrane CKI-3
- (52) *Iipili-t neri-neq qatsup-para.
 apple-PL:ABS eat-NEQ be.tired.of-1SG>3SG:IND
 Intended: ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-4

¹⁹The possessive inflections on examples (47)-(49) are anaphoric in the sense that they obligatorily reference the subject. As noted above, there is a second series of inflections for disjoint reference between the possessor and a higher subject. Disjoint possessive inflection on the *neq*-phrase is impossible in both the PNC and control constructions. For the PNC the impossibility of disjoint possessive morphology follows from the *neq*-phrase being verbal. For control structures it follows from the possessive morphology referencing the subject.

Instead such internal arguments are either incorporated (53) or realized as an oblique, e.g. an instrumental object of an antipassivized verb (54a) or an instrumental object of a notionally transitive, but formally intransitive verb form (54b); see also Fortescue (1984:45).

- (53) a. Puisi–tor–neq nuannara–arput.
 seal–consume–NEQ love–1PL>3SG:IND
 ‘We love to eat seal meat.’ Ellen Thrane BSY-2
- b. Iipili–tor–neq qatsup–para.
 apple–consume–NEQ be.tired.of–1SG>3SG:IND
 ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-6
- (54) a. Meeqqa–nik iku–i–neq nuannari–vara.
 child–INST.PL help–AP–NEQ love–1SG>3SG:IND
 ‘I love to help the children.’ Grethe Schmidt CHC-GS-2
- b. Iipili–nik neri–neq qatsup–para.
 apple–INST.PL eat–NEQ be.tired.of–1SG>3SG:IND
 ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-5

Prediction 4 If a control construction makes an oblique case available, the *neq*-phrase should be able to expone that case. We see this in the example in (55), where the matrix verb ‘be tired’ licenses a source argument. That source argument is realized with ablative case on the *neq*-phrase. Because the *neq*-phrase is not absolutive, the internal argument of *malisor*- ‘chase’ can be realized as an independent absolutive DP without running afoul of the single absolutive restriction. Compare (55) to (51) and (52).²⁰

- (55) Juuna–p qilalukka–t malisor–ner–mit qasu–voq.
 Juuna–ERG whale–PL:ABS chase–NEQ–ABL be.tired–3SG:IND
 ‘Juuna is tired from chasing whales.’ Bittner (1994:53), ex. 9a

Prediction 5 As an absolutive argument of a transitive verb, the *neq*-DP should control object agreement on the matrix control verb. As a nominalization, the ϕ -features of the *neq*-DP are 3SG (e.g. Hanink 2021:545-6, Picallo 2001). We thus correctly predict obligatory 3SG object agreement in control constructions.

- (56) Mersor–neq nuannari–**vara**.
 sew–NEQ love–1SG>3SG:IND
 ‘I love to sew.’ Ellen Thrane BTM-3

²⁰Carrier and Compton (2024) investigate a similar contrast in Inuktitut nominalizations involving a cognate suffix *-niq*. These *-niq* structures allow an internal absolutive argument when occurring in adjunct position, but not when occurring in argument position. Carrier and Compton (2024) attribute this contrast to a size difference between adjunct and argument *niq*-nominalizations. Adjunct nominalizations are large enough to license absolutive case, specifically by Force and Topic heads above *-niq*. Argument nominalizations lack the Force and Topic projections and hence absolutive case is not licensed. While we have not investigated the distribution of Force and Topic heads in Kalaallisut, it seems quite possible to us that Carrier and Compton’s analysis could extend to the Kalaallisut control constructions that we are comparing the PNC to. The *neq*-phrase is in an argument position (it is a complement to the control verb) and as such should lack Force and Topic projections without which no absolutive case is available for an internal argument of the embedded verb. The key point for us is that, in control constructions, *neq*/*niq*- are nominalizations in argument positions. That is what sets them apart for *neq*-phrases in the PNC.

- (57) *Mersor–neq nuannari–vunga.

sew–NEQ love–1SG:IND

Intended: ‘I love to sew.’

Ellen Thrane BTM-2

An alternative interpretation of (56) is that the 3SG object agreement is a form of default agreement. There are two reasons to resist that interpretation. First, coordinated *neq*-phrases can trigger 3PL object agreement:

- (58) Erinarsor–neq qitin–ner=lu nuannari–vakka
sing–NEQ dance–NEQ–CONJ love–1SG>3PL:IND

‘I love to sing and dance.’

Ellen Thrane DJE-5

This is direct evidence for a genuine number feature on the *neq*-phrase and unexplained on a default agreement analysis. Second, we don’t see default object agreement in other contexts where person-number features of a notional object are inaccessible for agreement. Recall that verbs only agree with ergative and absolutive arguments. As noted in the discussion of (54b) above, there is a class of notionally transitive verbs, including *neri*- ‘eat’, which occur in two case frames: ERG-ABS and ABS-INST.^{21,22}

In the ERG-ABS case frame, the verb shows transitive agreement:

- (59) Naja–p iipili–t pingasut neri–vai.
Naja–ERG apple–PL:ABS three:ABS eat–3SG>3PL:IND

‘Naja ate three apples.’

Grethe Schmidt CKL-GS-1

Since verbs can’t agree with oblique arguments, we might expect default 3SG object agreement in the ABS-INST frame. But that is impossible (60). Instead we find intransitive agreement (61).

- (60) *Naja iipili–nik pingasu–nik neri–vaa.
Naja:ABS apple–INST:PL three–INST:PL eat–3SG>3SG:IND

Intended: ‘Naja ate three apples.’

Grethe Schmidt CKL-GS-3

- (61) Naja iipili–nik pingasu–nik neri–voq.
Naja:ABS apple–INST:PL three–INST:PL eat–3SG:IND

‘Naja ate three apples.’

Grethe Schmidt CKL-GS-4

In other words, outside of control structures, there is no evidence for default agreement. In fact there is evidence against it from the ungrammaticality of (60). We conclude that control verbs do agree with their *neq*-phrase complement, lending support to our claim that the *neq*-phrase is nominal in these structures.

²¹ As pointed out by a reviewer, the ABS-INST construction in (54b) and (61) could also be analyzed as involving a null antipassive morpheme on the verb. Our argument against default case goes through under either analysis.

²² According to traditional treatments like those in Kleinschmidt (1851/1968), Bergsland (1955), Woodbury (1975), Fortescue (1984), and Sadock (1980), absolutive objects are definite and instrumental objects are indefinite. Bittner (1987) shows that the situation is considerably more complex and argues that the relevant difference is one of scope: absolutive objects take wide scope, whereas instrumental objects must take narrow scope. In our work there is not a consistent correlation between an absolutive case-marked object and a definite translation nor between an instrumental case-marked object and an indefinite translation. Since we are not concerned with semantic interpretation of the constructions under discussion, we set this issue aside and simply provide faithful English translations of the Danish translations established in elicitation.

5.2 Periphrastic negation structures

For the PNC, our hypothesis is that V-*neq* is a non-finite verbal projection, specifically a non-finite TP. From this hypothesis four predictions follow.

Prediction 1 As a TP, the *neq*-phrase cannot bear any case morphology, be it a possessive absolutive case (62), instrumental case (63), or allative case (64). (While not shown for reasons of space the other five cases—ablative, equative, ergative, locative, and prolative—are equally impossible on the *neq*-phrase in the PNC.)

- (62) *Naja-p mersor-ni-ni ajor-paa.
 Naja-ERG sew-AP-4SG>3SG:ABS HAB:NEG-3SG>3SG:IND
 Intended: ‘Naja loves to sew.’ Ellen Thrane CII-7
- (63) *Naja biiler-nim-mik ajor-poq.
 Naja:ABS drive-NEQ-INST HAB:NEG-3SG:IND
 Intended: ‘Naja doesn’t drive.’ Grethe Schmidt BRF-GS-4
- (64) *Meeqqa-t ikior-nim-mut ajor-pakka.
 child-PL:ABS help-NEQ-ALL HAB:NEG-1SG>3PL:IND
 Intended: ‘I didn’t help the children.’ Grethe Schmidt BRG-GS-3

Absolutive and instrumental case are the most obvious cases to consider. Absolutive is relevant because the *neq*-phrase is the complement of a finite verbal form, as is the case for absolutive DPs. Instrumental is relevant because it alternates with absolutive case in antipassive constructions yielding intransitive agreement on the verb, which is associated with an intransitive *neq*-complement to *ajor*-, here ‘drive’. The allative case example in (64) is also ungrammatical, as are all other oblique cases.

Prediction 2 Without an absolutive internal argument of the lexical verb, the subject cannot be ergative, whether the negative auxiliary shows transitive or intransitive agreement:

- (65) *Naja-p biiler-neq ajor-paa/poq.
 Naja-ERG drive-NEQ HAB:NEG-3SG:IND/3SG:IND
 Intended: ‘Naja doesn’t drive.’ Grethe Schmidt CIH-GS-4/CIH-GS-3

Prediction 3 Since the *neq*-phrase is not absolutive, absolutive case is available for internal arguments of the lexical verb, as seen for *meeqqa-ni* ‘her children’ in (66).

- (66) Naja-p meeqqa-ni ikior-neq ajor-pai.
 Naja-ERG child-4SG>3SG:ABS help-NEQ HAB:NEG-3SG>3PL:IND
 ‘Naja_i didn’t help her_i children.’ Grethe Schmidt BRJ-GS-2

Prediction 4 As the *neq*-phrase is not absolutive, it cannot control object agreement on the negative verb:

- (67) *Biiler–neq ajor–para.
 drive–NEQ HAB:NEG–1SG>3SG:IND
 Intended: ‘I don’t drive.’

Grethe Schmidt CHA-GS-2

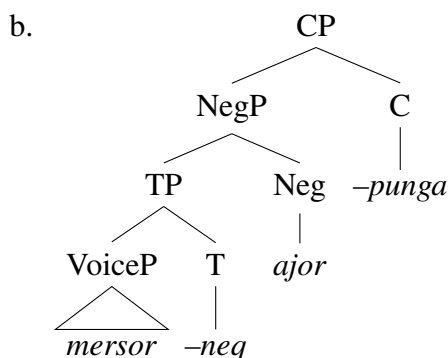
To summarize: in the PNC the *neq*-phrase is syntactically inert as regards case and agreement. It does not bear case itself, it does not license ergative case on the higher subject, it is transparent for absolutive case assignment, and it does not control agreement. This follows straightforwardly on the hypothesis that in the PNC the *neq*-phrase is exclusively verbal.

6 The structure of *neq*- constructions

In this section we provide phrase-structural analyses of the PNC and control structures, based on the observations made above. As the analysis revolves around the syntactic status of the *neq*-suffix, we build our analysis from an examination of the position of *-neq* relative to the position of argument structure morphology and inflectional morphology. We show that in both the PNC and control structures the *neq*-head is positioned between argument structural projections (labeled VoiceP in (68b) and (69b)) and inflectional heads (section 6.1). For control verbs, nothing more needs to be said. For the PNC, we make the further argument that *ajor-* is a Neg head positioned above T (section 6.2). The two analyses are illustrated in the trees below; following Compton (2018), we take mood and agreement to be exponed in C.

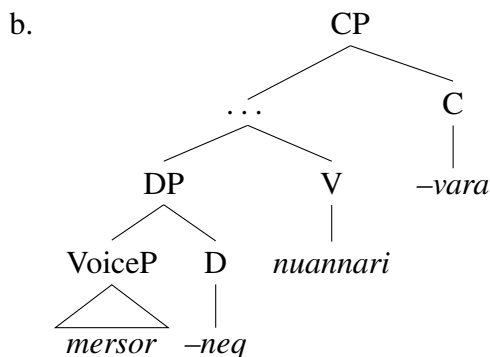
- (68) a. Mersor–neq ajor–punga.
 sew–NEQ HAB:NEG–1SG:IND
 ‘I don’t sew.’

Ellen Thrane CJP-4



- (69) a. Mersor–neq nuannari–vara.
 sew–NEQ love–1SG>3SG:IND
 ‘I love to sew.’

Grethe Schmidt BTR-GS-1



6.1 Structure below *-neq*

Before we document the syntactic material that *-neq* can embed we need to say a few words about the clausal structure we assume for Kalaallisut and about the scope of our investigation. Morphology reveals that the following heads can appear on the verbal spine: i) V which corresponds to the verbal root and introduces any internal arguments; ii) Caus which introduces a causer argument and demotes the embedded Agent to oblique, iii) Appl which promotes an oblique argument to direct object; iv) the antipassive head *ap* (Yuan 2018:205ff), which demotes the notional object to an oblique, and v) Voice, which introduces the Agent in active clauses and hosts the passive suffix in passive clauses. We assume that these heads are only present when warranted by syntactic material in a particular construction. That is, we are not adopting a cartographic approach. We will not be concerned with the relative height of these heads, but simply document that any and all of them can occur inside the *neq*-phrase. (For ordering of the corresponding derivational morphemes see Fortescue (1980).) We also limit ourselves to argument-structure changing morphology and indisputably inflectional morphology, which is mood and agreement. We do this for two reasons. First, there are 300–400 derivational suffixes that occur in Kalaallisut verbs (Fortescue 1980:259) and it would be practically impossible to examine for each of them whether it can occur inside the *neq*-phrase. Second, the suffixes that we choose to look at are exactly the ones that are most informative for our analytic purpose. Argument-changing morphology interacts with case and agreement, which is what we are analyzing. Inflectional morphology is hosted in C and the very outermost morphology of the verb word. As such, it let's us determine whether the upper bound of *-neq* is below the finite CP.

Our documentation relies on the Mirror Principle which states that “[m]orphological derivations directly reflect syntactic derivations (and vice versa)” (Baker 1985:375). Of particular relevance is its corollary regarding morpheme order and the hierarchy of syntactic projections: if morpheme α occurs closer to the root than morpheme β , then the syntactic head responsible for α is lower in the syntactic structure than the syntactic head responsible for β . We will see that any argument-structure changing morphology can occur closer to the root than *-neq*, which indicates that VP, *ap*P, CausP, ApplP and VoiceP can all be projected below the *neq*-head.

In the PNC this is true in all circumstances, whereas it is modulated by the ban on multiple absolutes in control structures. In neither construction is inflectional morphology inside *-neq* possible.

We start our examination with basic active clauses. As seen above already, these are licit in both the PNC and in control structures.

(70) ACTIVE VOICE

- a. Naja-p meeqqa-t ikior-neq ajor-pai
Naja-ERG child-PL:ABS help-NEQ HAB:NEG-3SG>3PL:IND

‘Naja didn’t help the children.’

Grethe Schmidt BRI-GS-2

- b. Naja-p mersor-neq nuannari-vaa.
Naja-ERG sew-NEQ love-3SG>3SG:IND

‘Naja loves to sew.’

Grethe Schmidt CIB-GS-2

neq-phrases formed on passives are also possible in both constructions, with the passive suffix *-neqar* occurring inside *-neq* :

(71) PASSIVE VOICE

- a. Meeqqa-t Naja-mik ikior-**neqar**-neq ajor-put.
child-PL:ABS Naja-INST help-PASS-NEQ HAB:NEG-3PL:IND

‘The children are not being helped by Naja.’

Ellen Thrane BRN-7

- b. Meeqqa-t Naja-mik ikior-**neqar**-neq nuannari-vaat.
child-PL:ABS Naja-INST help-PASS-NEQ love-3PL>3SG/PL:IND

‘The children love to be helped by Naja.’

Ellen Thrane CIP-1

The antipassive morpheme *-i* likewise occurs inside *-neq* indicating that an antipassive phrase, apP, can also be contained in the *neq*-phrase:

(72) ANTIPASSIVE

- a. Naja meeqqa-nik ikiu-**i**-neq ajor-poq.
Naja:ABS child-INST:PL help-AP-NEQ HAB:NEG-3SG:IND

‘Naja doesn’t help the children.’

Ellen Thrane BRP-1

- b. Naja-p meeqqa-nik ikiu-**i**-neq nuannari-vaa.
Naja-ERG child-INST:PL help-AP-NEQ love-3PL>3SG:IND

‘Naja loves to help the children.’

Ellen Thrane CIP-3

The causative suffix *-tit* can occur inside *-neq* in the PNC (73).²³

- (73) Juuna-p Naja-mut meeqqa-t ikior-**tin**-neq ajor-pai.
Juuna-ERG Naja-ALL child-PL:ABS help-CAUS-NEQ HAB:NEG-3SG>3PL:IND

‘Juuna doesn’t let Naja help the children.’

Ellen Thrane CIK-1

In contrast, causative *-tit* inside *-neq* is ungrammatical in the control example in (74).

- (74) *Juuna-p Naja-mut meeqqa-t ikior-**tin**-neq nuannari-vaa.
Juuna-ERG Naja-ALL child-PL:ABS help-CAUS-NEQ love-3SG>3SG:IND

Intended: ‘Juuna loves letting Naja help the children.’

Ellen Thrane CFK-11

²³As discussed by Fortescue (1984:84ff), *-tit* allows a reading akin to an analytic ‘let’-causative, as in (73). Also note that the final consonant of the causative morpheme assimilates to the initial consonant of *-neq* producing the surface form *-tin-neq*. Here we use *-tit* as the citation form following Fortescue (1984:84)

We attribute this contrast to the single absolutive restriction: in (73) there is only one absolutive DP, namely the *meeqqa-t* ‘children’. In (74) there are two: *meeqqa-t* and the nominalized DP that contains it ([_{DP} *Naja-mut meeqqa-t ikior-tin-neq*] ‘letting Naja help the children’). Evidence that it is indeed the single absolutive restriction that is at play and not a difference in whether the *neq*-phrase can embed CausP in the two constructions comes from the example in (75). This is an antipassivized version of (74). As such, the theme is not absolutive, but instrumental and the entire construction contains only one absolutive, namely the DP complement of the control verb. The grammaticality of (75) thus confirms that the *neq*-head can embed CausP in a control construction, just as it can in the PNC.

- (75) Juuna-p Naja-mut meeqqa-nik iku-i-**tin-neq** nuannari-vaa.
 Juuna-ERG Naja-ALL child-INST.PL help-AP-CAUS-NEQ love-3SG>3SG:IND
 ‘Juuna loves letting Naja help the children.’ Ellen Thrane CKP-1

The applicative morpheme *-ut* has multiple functions (Woodbury 1977), two of which are relevant here.²⁴ First, it transitivizes an intransitive verb (76). Second, when applied to a ditransitive verb, it promotes an oblique theme to direct object, as seen in (77b). (The underived form in (77a) shows that Kalallisuut has secundative alignment in basic ditransitive structures.)

- (76) TRANSITIVIZING FUNCTION OF -UT
- a. Ummi-vunga.
 aim-1SG:IND
 ‘I took aim.’ Ellen Thrane CKQ-1
- b. Orpi-it ummi-**up**-pakka.
 tree-PL:ABS aim-APPL-1SG>3PL:IND
 ‘I aimed for the trees.’ Ellen Thrane CKQ-2
- (77) THEME PROMOTING FUNCTION OF -UT
- a. Meeqqa-t poortukka-nik nassip-pakka.
 child-PL:ABS package-INST.PL send-1SG>3PL:IND
 ‘I sent the children the packages.’ Ellen Thrane CIM-1
- b. Meeqqa-nut poortukka-t nassi-**up**-pakka.
 child-PL:ALL package-PL:ABS send-APPL-1SG>3PL:IND
 ‘I sent the packages to the children.’ Ellen Thrane CIM-2

The PNC allows both applicative structures to surface inside the *neq*-phrase:

- (78) a. Orpi-it ummi-**un-neq** ajor-pakka.
 tree-PL:ABS aim-APPL-NEQ HAB:NEG-3SG>3PL:IND
 ‘I don’t aim for the trees.’ Ellen Thrane CKQ-4
- b. Meeqqa-nut poortukka-t nassi-**un-neq** ajor-pakka.
 child-PL:ALL package-PL:ABS send-APPL-NEQ HAB:NEG-1SG>3PL:IND
 ‘I didn’t send the package to the children.’ Grethe Schmidt CJM-GS-1

²⁴Like causative *-tit*, the final consonant of *-ut* assimilates to a following consonant.

In the control structure, we again see the effect of the single absolutive restriction. Specifically, the structures in (79) are ungrammatical because in addition to the absolutive *neq*-DP, there is a second absolutive DP (*orpiit* in (79a) and *poortukkat* in (79b)).

- (79) a. *Orpi-it ummi-**un-neq** qatsup-para.
 tree-PL:ABS aim-APPL-NEQ be.tired.of-1SG>3SG:IND
 Intended: ‘I am tired of aiming for the trees.’ Ellen Thrane CKR-2
- b. *Meeqqa-nut poortukka-t nassi-**un-neq** nuannari-vara.
 child-PL:ALL package-PL:ABS send-APPL-NEQ love-1SG>3SG:IND.
 Intended: ‘I love to send packages to the children.’ Ellen Thrane CKI-3

Again we can show that the issue is not embedding ApplP under the *neq*-head, but instead the ban against multiple absolutives. Starting with the ungrammatical applicative structure in (79a), we can avoid multiple absolutives by antipassivizing the verb so that the theme is realized with instrumental case. This strategy is confirmed by the grammaticality of (80a). Similarly, we can antipassivize the ditransitive verb in (79b) to produce the grammatical (80b); the final /p/ of the causative suffix assimilates to the initial /s/ of the antipassive morpheme.

- (80) a. Orpin-nik ummi-**us-si-neq** qatsup-para.
 tree-INST:PL aim-APPL-AP-NEQ be.tired.of-1SG>3SG:IND
 ‘I am tired of aiming for the trees.’ Ellen Thrane CKR-8
- b. Meeqqa-nut poortukka-nik nassi-**us-si-neq** nuannari-vara.
 child-PL:ALL package-INST:PL send-APPL-AP-NEQ love-1SG>3SG:IND.
 ‘I love to send packages to the children.’ Ellen Thrane CLO-3

In summary, the *neq*-phrase can embed active and passive VoiceP, ApplP, CausP and apP.

In contrast *neq*-phrases verbs cannot contain inflectional morphology, specifically mood and agreement morphology, in neither the PNC nor control structures. The most likely mood to be possible is contemporative mood, which is used when the subject of an embedded clause is coreferential with the matrix subject, as in (81).

- (81) Naja_i oqar-poq [_{CP} pro_i tamatigut angerla-jaar-**luni**].
 Naja:ABS say-3SG:IND 3SG:ABS always leave-early-4SG:CONT
 ‘Naja_i said that she_i always leaves early.’ Ellen Thrane BCC-2

In control constructions the embedded subject is likewise coreferential with the matrix subject (Naja_i loves PRO_i to drive) and thus if any mood is available it would be the contemporative mood. Yet, contemporative mood inside the *neq*-phrase in a control construction is impossible as (82) shows.

- (82) *Naja-p biiler-**luni-neq** nuannari-vaa.
 Naja-ERG drive-4SG:CONT-NEQ:ABS love-1SG>3SG:IND
 Intended: ‘Naja loves to drive.’ Ellen Thrane CIR-4

Contemporative mood morphology inside the *neq*-phrase in the PNC is similarly ungrammatical:

- (83) *Naja biiler-**luni**-neq ajor-poq.
 Naja:ABS drive-4SG:CONT-NEQ HAB:NEG-3SG:IND
 Intended: ‘Naja doesn’t drive.’ Ellen Thrane CIR-2

Unsurprisingly, the other seven moods—indicative, interrogative, optative, participial, causative, conditional, imperative—are also impossible inside *-neq* in both control structures and the PNC. Here we illustrate with indicative (84) and participial moods (85).

- (84) a. *Naja biiler-**por**-neq ajor-poq.
 Naja:ABS drive-3SG:IND-NEQ HAB:NEG-3SG:IND
 Intended: ‘Naja doesn’t drive.’ Ellen Thrane CIR-1
- b. *Naja-p biiler-**por**-neq nuannari-vaa.
 Naja-ERG drive-3SG:IND-NEQ love-1SG>3SG:IND
 Intended: ‘Naja loves to drive.’ Grethe Schmidt DAM-GS-2
- (85) a. *Naja biiler-**tor**-neq ajor-poq.
 Naja:ABS drive-3SG:PART-NEQ HAB:NEG-3SG:IND
 Intended: ‘Naja doesn’t drive.’ Grethe Schmidt DAL-GS-1
- b. *Naja-p biiler-**tor**-neq nuannari-vaa.
 Naja-ERG drive-3SG:PART-NEQ:ABS love-1SG>3SG:IND
 Intended: ‘Naja loves to drive.’ Grethe Schmidt DAL-GS-3

The data provided in this section are summarized in Table 1. Recall that for the causative and applicatives, the control construction requires antipassivization to not run afoul of the single absolutive restriction.

STRUCTURE	PNC	CONTROL
active	✓	✓
passive	✓	✓
antipassive	✓	✓
applicative of intransitive	✓	✓
applicative of transitive	✓	✓
causative	✓	✓
agreement/mood	✗	✗

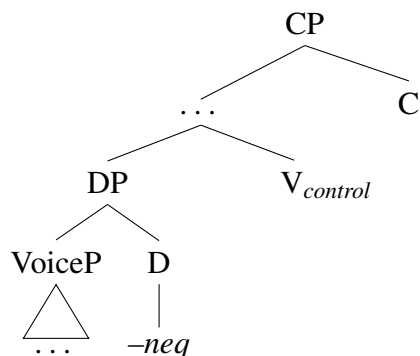
Table 1: Possible structure below *-neq*

On our structural analyses of the PNC and the control construction, this distribution is due to the *neq*-head invariably taking VoiceP as its complement and never CP, as would be required for mood and agreement morphology to occur inside *-neq*. In section 7, we return to the observation that the *neq*-head embeds the same amount of structure in the two constructions and suggest that it stems from the PNC having developed from a control structure.

6.2 Structure above the *neq*-phrase

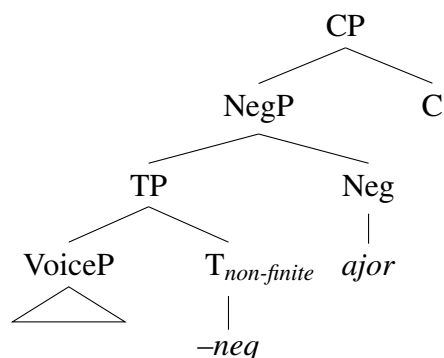
In control structures *-neq* heads a DP, which in turn occupies the complement position of the control verb.

(86) CONTROL



In the PNC, the *neq*-phrase is a non-finite TP complement of the negative head *ajor*-. Following Holmberg et al.'s (1993:183–185) analysis of negative auxiliaries in Finnish (ISO 639: fin), we propose that *ajor*- projects a NegP as in (87).²⁵

(87) PNC



Evidence for a NegP analysis comes from the incompatibility of the negative auxiliary and suffixal negation *-nngit*:²⁶

- (88) *Naja biiler-neq **aju-nngi**-laq.
 Naja:ABS drive.car-NEG HAB:NEG-neg-3SG:IND:NEG
 Intended: 'Naja drives.' (lit. 'Naja doesn't not drive')

Grethe Schmidt BSR-GS-4

We note that the lexical verb *ajor*- 'be bad' can co-occur with suffixal negation as in the example in (89).

²⁵This analysis does not, in and of itself, explain the habitual meaning of the PNC. A reviewer suggests that *ajor*- could be a single morpheme realization of a span of habitual Asp and Neg (Svenonius 2012). We find this idea appealing, but have chosen to leave it for future research, partly because we have not investigated whether Asp and Neg form a span in the PNC, partly for reasons of space.

²⁶There are three morpho-phonological processes at play in (88). First the negative suffix deletes the final /t/ of the negative auxiliary. This in turn bleeds the phonological rule that lowers /u/ to /o/ before /t/. Third *-nngit* triggers the special negative allomorph of the 3SG:IND morpheme, *-laq* in place of the regular form *-poq*.

- (89) pisiari–gukku **aju–ssa–nngi–laq**
 buy–2SG>3SG:COND be.bad–FUT–NEG–3SG:IND:NEG
 ‘It would be good if you bought it.’ Fortescue (1984:33)

In accord with the ungrammaticality judgment in (88), we have not found any examples of negative auxiliary *ajor* and *-nngit* cooccurring in Bittner’s text collection nor in Fortescue (1984), Sadock (2003), or Kahn and Valijärvi (2022). In addition to being in complementary distribution, both negative morphemes are followed by mood and agreement morphology as seen in (90).

- (90) a. Biili–**nngi–langa**.
 drive–NEG–1SG:IND:NEG
 ‘I don’t drive.’ Grethe Schmidt CMB-GS-2
- b. Biiler–neq **ajor–punga**.
 drive–NEQ HAB:NEG–1SG:IND
 ‘I don’t drive.’ Grethe Schmidt CMB-GS-3

The proposed structure also explains the ungrammaticality of (91):

- (91) *Naja biiler–neq **aju–ssa–aq**.
 Naja drive.car–NEQ HAB:NEG–FUT–3SG:IND
 Intended: ‘Naja will not drive in the future.’ Grethe Schmidt BSF-GS-1

In this example future tense is stacked on top of *ajor-* which doesn’t fit the structure in (87), where T is below Neg.

Moreover, in its habitual negation use, *ajor-* does not stand in a thematic relationship to any of the DPs it co-occurs with—i.e. *ajor-* does not assign any theta roles—suggesting that it is not a lexical verb. Finally, the fact that *ajor-* contributes negation makes it a good fit for Neg⁰. Our conclusion is therefore that *ajor-* projects a NegP, as in (87).

7 Diachrony

Above we have argued that the PNC differs from control constructions in two key respects: the *neq*-phrase is verbal in the PNC, but nominal in control constructions. Second, *ajor-* is an auxiliary, whereas control verbs like *nuannari-* (‘love to’) are lexical verbs. In this section we sketch a diachronic analysis of how these differences came to be. In short we propose that the auxiliary *ajor-* developed from a lexical control verb *ajor-* in tandem with a change in the category of the *-neq* phrase from nominal to verbal. Both are well-established pathways of change. Heine and Kuteva (2002) and Heine and Kuteva (2019) list numerous examples of lexical verbs developing into auxiliaries, including negative and habitual ones, from a wide range of languages. The change from a nominalized constituent to a verbal one is documented in Gildea (2008) and Malchukov (2019). The confluence of these two changes are schematized in a general way in (92), where AFF stands for some affix, and the schema in (93) illustrated the specific change we propose for Kalaallisut.

- (92) [V-AFF]_{nominal} V_{lexical} ⇒ [V-AFF]_{verbal} Aux

(93) $[V\text{-}neq]_{DP} \text{ ajor}_{control} \Rightarrow [V\text{-}neq]_{TP} \text{ ajor}_{aux}$

We offer six observations in support of the hypothesized syntactic change in (93).

1. *Ajor-* as a control verb. Various dictionaries include a transitive entry for the verb *ajor-* along with an entry of its auxiliary usage. The Danish–Kalaallisut dictionary DAKA (<https://iserasuaat.gl/daka/>) lists the two entries in (94):²⁷

- (94) a. *ajorpaa*: ‘klarer det ikke, magter det ikke’
 b. *ajorpoq*: *pujortarneq ajorpoq* ‘ryger ikke’

These entries are translated in (95).

- (95) a. *ajorpaa*: ‘can’t do it, doesn’t have the capacity to’
 b. *ajorpoq*: *pujortarneq ajorpoq* ‘does not smoke’

(95a) shows transitive agreement (*-paa* 3SG>3SG:IND), while (95b) shows intransitive agreement (*-poq* 3SG:IND). The meaning of transitive *ajor-* is similar to that of the control verbs *saper-* ‘can’t manage’ and *artor-* ‘be unable’. This, together with its transitive agreement, suggests that transitive *ajor-* in (94a) and (95a) is also a control verb. None of the four Kalaallisut speakers we consulted recognize the transitive usage of *ajor-* in (94a)/(95a), including co-authors Grethe Schmidt and Ellen Thrane. We take this to suggest that for these speakers the grammaticalization schematized in (96) is completed and the original transitive control construction has been lost.

(96) $[V\text{-}neq]_{DP} \text{ ajor}_{control} \Rightarrow [V\text{-}neq]_{TP} \text{ ajor}_{aux}$

2. Comparative evidence. The Comparative Eskimo Dictionary lists the Proto-Inuit root in (97) (Fortescue et al. 1994:60).

- (97) **ayuuq-** ‘be unable to reach or do’²⁸

Cognates of this verb root with similar meanings are observed in four modern varieties of Inuit (Seward Peninsular Inuit, North Alaskan Inuit, Western Canadian Inuit, and Eastern Canadian Inuit) and possibly in the Alutiiq Alaskan variety of Yupik (Fortescue et al. 1994:60). However, aside from Kalaallisut, no modern Inuit variety employs the cognates of *ajor-* as a negative auxiliary (Richard Compton, p.c.). We take this to support our view that the PNC is an innovation in Kalaallisut.

3. General function of *-neq* as a nominalizer. Further support for the PNC being an innovation comes from the fact that the overall distribution of *-neq* in Kalaallisut is as a nominalizer (Fortescue 1984:44ff), making its verbal use in the PNC the odd one out. While we cannot give a full analysis of the nominalizing functions of *-neq*, we include the examples from Fortescue (1984) below to illustrate its range. Note that Fortescue glosses *-neq* with NOM for nominalizer. Also note that Fortescue uses the spelling *-niq/nir*, not the standard spelling *-neq/ner*.

²⁷DAKA is an online dictionary comprising two dictionaries: *Oqaatsit* (1997) and *Ordbogen* (2003).

²⁸In the modern Kalaallisut orthography, this would be represented as *ajoq-*, where the /q/ alternates with the /r/ when it precedes a suffix that begins with a consonant.

- (98) piuma-sa-qar-**nir**-anit angi-niru-suq.
want-PASS:PART-have-NOM-3SG>3SG:ABL be.big-more-INTRANS:PART
'bigger than he wanted' Fortescue (1984:103)
- (99) atuarvi-qar-**nir**-mik inatsit
school-have-NOM-INST law
'the law pertaining to schools' Fortescue (1984:112)
- (100) nipi-sia-ssa-tua-a-sara-aq timmia-nngu-it
sound-received-FUT-only-COP-HAB-3SG:IND bird-small-PL:ERG
 qarlur-**nir**-at.
 twitter-NOM-3PL>3SG:ABS
'The only thing to be heard would be the twitter of little birds.'
Fortescue (1984:183)

The fact that *-neq* functions as a nominalizer elsewhere supports our hypothesis that its nominalizing use in control constructions represents the original function of *-neq* and that its verbal use in the PNC is a localized innovation.

4. Grammaticalization of other negative control verbs. In descriptive works, the auxiliary *ajor-* is grouped with three other ‘negative verbs’ (Fortescue 1984:47–48, Kahn and Valijärvi 2022:245–247, and Sadock 2003:10–11,33): *artor-* ‘to be unable’, *nalu-* ‘to not know’, and *saper-* ‘to not manage (to do something)’.

Of these, *nalu-* and *saper-* can participate in PNC syntax, as observed in (101) and (102).

- (101) Naja biiler-neq **nalu-voq**.
 Naja:ABS drive-NEQ not.know-3SG:IND
 ‘Naja doesn’t know how to drive.’ Ellen Thrane BPA-2
- (102) Naja mersor-neq **saper-poq**.
 Naja:ABS sew-NEQ not.manage-3SG:IND
 ‘Naja can’t sew.’ Ellen Thrane CIT-1

These examples display PNC syntax in that the subject is in absolutive case (because the lexical verb is intransitive) and the agreement suffix of the negative verb is intransitive. In contrast, *artor-* only allows the syntax of control verbs, in which the subject has ergative case and the agreement suffix is transitive:

- (103) Naja-p mensor-neq artor-paa.
Naja-ERG sew-NEQ be.unable-3SG>3SG:IND
'Naja is unable to sew.'
Ellen Thrane CKX-2
- (104) *Naja mensor-neq artor-poq.
Naja:ABS sew-NEQ be.unable-3SG:IND
Intended: 'Naja is unable to sew.'
Ellen Thrane CIT-2

We can make sense of this pattern if the four negative verbs (*ajor-*, *artor-*, *nalu-*, *saper-*) originated as control verbs (cf. Fortescue et al. 1994:212) and three of them (*ajor-*, *nalu-*, *saper-*) evolved to become negative auxiliaries. In other words, there is a more general grammaticalization pathway from control verb to auxiliary in Kalaallisut.

5. Variable control-PNC syntax. Whereas *ajor-* has lost its meaning as a control verb and thus its ability to participate in control verb syntax, the negative auxiliaries (*nalu-* and *saper-*) preserve control verb syntax (105a and 106a) but may alternatively participate in PNC syntax (105b and 106b).

- (105) a. Naja-p biiler-neq nalu-vaa.
 Naja-ERG drive-NEQ not.know-3SG>3SG:IND
 ‘Naja doesn’t know how to drive.’ Ellen Thrane CIX-3
- b. Naja biiler-neq nalu-voq.
 Naja:ABS drive-NEQ not.know-3SG:IND
 ‘Naja doesn’t know how to drive.’ Ellen Thrane BPA-2
- (106) a. Naja-p mersor-neq saper-paa.
 Naja-ERG sew-NEQ not.manage-3SG>3SG:IND
 ‘Naja can’t sew.’ Ellen Thrane CLP-4
- b. Naja mersor-neq saper-poq.
 Naja:ABS sew-NEQ not.manage-3SG:IND
 ‘Naja can’t sew.’ Ellen Thrane CIT-1

The stages of development along the path to negative auxiliaries (PNC syntax) are indicated for each verb in Table 2.

	CONTROL	PNC
<i>artor-</i> ‘be unable’	✓	✗
<i>saper-</i> ‘not manage’	✓	✓
<i>nalu-</i> ‘not know’	✓	✓
<i>ajor-</i> HAB:NEG	✗	✓

Table 2: Degree of grammaticalization of the negative verbs

Following our proposal, the status of the verbs is understood as follows. One verb, *artor-*, has not participated in the change from control verb to auxiliary verb. Two verbs, *nalu-* and *saper-*, have begun the change, and one verb, *ajor-*, has completed the change.

6. Internal structure of *neq*-phrase. As established in section 6, PNC and control constructions exhibit identical restrictions on syntactic structure inside the *neq*-phrase. Under the hypothesized change from a negative control construction to the PNC, this follows from the Inertia Principle, which states that “syntactic change should not arise, unless it can be shown to be *caused*.” (Longo-bardi 2001:283; see also Keenan 2002, and Roberts 2017). The differences in case and agreement between control constructions and the PNC are examples of caused changes. Case and agreement depend directly on syntactic category: nominals require case and control agreement, verbs do not occur with case and do not control agreement. Thus, when a syntactic phrase changes category from a nominal element to a verbal element, this causes changes to the distribution of case and agreement. The differences in case and agreement documented in section 5 are the synchronic consequences of this change. On the other hand, nothing forces the amount of structure possible

below *-neq* to change due to the category change. Therefore, by the Inertia Principle, it does not change. Based on this reasoning, we claim that the fact that the restrictions on the structure internal to the *-neq* phrase are exactly the same in control constructions follows from the proposed change and therefore counts as an argument in its favor.

Semantic change. In the previous section we referenced the Inertia Principle, which states that syntactic change must be caused. On our proposal, that immediately raises a question: what caused the *-neq* phrase to change from a nominal element to a verbal element and, relatedly, what caused *ajor-* to change from a verb to an auxiliary? While we cannot offer proof, we would like to suggest that it was a semantic change that caused these syntactic changes. In particular, a semantic change from a modal negative meaning of *ajor-* ‘be unable to reach or do’ to just the negative meaning ‘not do’. This would be an instance of semantic bleaching, “whereby a more complex meaning is reduced to its nucleus” (Heine and Kuteva 2019:295). This semantic change, we hypothesize, caused a syntactic category change of *ajor-* from V to Aux, which in turn caused a syntactic change in the syntactic status of the *-neq* from a nominal to verbal, as auxiliaries take verbal complements not nominal ones. This is admittedly speculative, but does, we think, add up to a coherent hypothesis about the diachrony of the PNC.

8 Conclusion and open issues

In this paper we have described and analyzed a periphrastic negation construction in Kalaallisut by way of a systematic comparison with a superficially similar control construction. Both constructions involve an embedded non-finite verb combined with the suffix *-neq*, but we have argued that the syntactic category of this suffix differs in the two constructions. In control constructions it is a nominalizing D head; in the PNC it is a non-finite T head. We have shown that this one difference accounts for a host of differences between the two constructions as regards case, agreement, word order, possessive morphology and complementation. The second part of the PNC is a finite negative auxiliary *ajor-* which we analyzed as a Neg head situated between T and C. This analysis explains why *ajor-* does not assign a theta-role (it is not a lexical verb); why it does not inflect for tense (it is above tense); why it does not co-occur with suffixal negation (the two morphemes occupy the same position).

In addition to this synchronic analysis we sketched a diachronic analysis according to which the PNC developed from a control structure by two interlocking processes: the *neq*-nominalization was reanalyzed as a non-finite TP and the negative control verb *ajor-* ‘can’t manage’ was grammaticalized into an auxiliary. In support of this analysis, we showed that other verbs designated ‘negative verbs’ in the descriptive and pedagogical literature are at various stages of the same change, though they all retain their control usage. Interestingly, there is evidence that some non-negative control verbs might be heading down the same path. Consider the example in (107):

- (107) Iipili-t neri-neq qatsup-pakka.
 apples-PL:ABS eat-NEQ be.tired.of-1SG>3PL:IND
 ‘I am tired of eating apples.’

Grethe Schmidt CZO-GS-5

Here the finite verb agrees with the third person plural absolutive internal argument of the embedded verb (*iipilit* ‘apples’), not with the *neq*-phrase. This is different from the regular control construc-

tion where the control verb shows third person singular object agreement with its nominalized complement.

- (108) Iipili–tor–neq qatsup–**para**.
 apple–consume–NEQ be.tired.of–1SG>3SG:IND
 ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-6

In fact, (107) is parallel to the PNC example in (109), where the negative auxiliary shows third person plural object agreement with the internal argument of ‘eat’.

- (109) Iipili–t neri–neq ajor–**pakka**.
 apple–PL:ABS eat–NEQ HAB:NEG–1SG>3PL:IND
 ‘I don’t eat apples.’ Grethe Schmidt DAM-GS-1

We tentatively suggest that the construction in (107) also involves reanalysis of the *neq*-phrase as a verbal projection (a non-finite TP) and restructuring of a biclausal control construction into a monoclausal structure. More work is needed to substantiate this analysis, but we can point to one piece of evidence in its favor, namely that it accounts for the contrast between (107) and (110).

- (110) *Iipili–t neri–neq qatsup–para.
 apple–PL:ABS eat–NEQ be.tired.of–1SG>3SG:IND
 Intended: ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-4

In (110), *qatsup*- ‘be tired of’ shows third person singular object agreement. As argued at length in section 5.1, this is object agreement with the *neq*-phrase. In that structure an absolutive internal argument cannot be case licensed, because the unique absolutive case is assigned to the *neq*-phrase. However, in (107) an internal absolutive argument can be case licensed because the *neq*-phrase is a verbal projection and not in need of case.

As mentioned in footnote 20, Carrier and Compton (2024) analyze absolutive case assignment in nominalization structures formed with the Inuktitut suffix *-niq*, which is cognate with Kalaallisut *-neq*. A second avenue of future research is therefore to compare and possibly integrate our work with theirs.

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Appendix

This appendix consists of two tables compiled by Arly Madeline Siu from the original sources for WALS 112A (Negative Morphemes) for value "Negative auxiliary verb". There are 47 languages with this value in the WALS sample. The first table (Table 3) adds two pieces of information for each language: its morphological type (agglutinating, fusional, isolating, or polysynthetic) and whether it is reported to have a negative affix in addition to its negative auxiliary. A hyphen '-' indicates that we could not verify whether the language has a negative affix because the source was not available to us. These two added pieces of information help to further appreciate the typological status of the Kalaallisut negation system (polysynthetic with a negative affix in addition to a negative auxiliary).

The second table (Table 4) records the iso code, genetic affiliation, and source publication for each of the 47 languages.

Language Name	Morphological Type	Negative Affix
Abui	Agglutinating & polysynthetic	No
Achumawi	Agglutinating	No
Agta (Central)	Agglutinating	Yes
Armenian (Western)	Agglutinating	No
Avar	Agglutinating	-
Bontok	Agglutinating	No
Didinga	Agglutinating	No
Diegueño (Mesa Grande)	Polysynthetic	No
Dinka	Agglutinating	No
Enets	Agglutinating	No
Finnish	Agglutinating	No
Futuna-Aniwa	Agglutinating	No
Grebo	Agglutinating	No
Halkomelem (Upriver)	Polysynthetic	No
Hualapai	Polysynthetic	No
Ignaciano	Polysynthetic	No
Katcha	Agglutinating	No
Koegu	Agglutinating	Yes
Kolami	Agglutinating	Yes
Kombio	Not polysynthetic	-
Komi-Permyak	Agglutinating	No
Komi-Zyrian	Agglutinating	No
Komo	Agglutinating	No
Kyuquot	Polysynthetic	Yes
Majang	Not polysynthetic	-
Mano	Agglutinating	No
Mari (Meadow)	Polysynthetic	Yes
Moro	Agglutinating	No
Naga (Mao)	Polysynthetic	No
Nahali	Not polysynthetic	-
Nakanai	Agglutinating	No
Nenets	Polysynthetic	No
Nganasan	Polysynthetic	No
Nisgha	Polysynthetic	No
Pokot	Polysynthetic	Yes
Quileute	Polysynthetic	No
Rukai (Tanan)	Agglutinating	Yes
Saami (Northern)	Fusional	No
Sanuma	Agglutinating	No
Selknam	Polysynthetic	Unclear
Siroi	Agglutinating	No

Squamish	Polysynthetic	No
Taiof	Agglutinating	No
Thai	Isolating	No
Tsez	Polysynthetic	-
Udihe	Agglutinating	No
Zoque (Copainalá)	Agglutinating	No

Table 3: Typology of Negative Auxiliaries

Language Name	ISO	Language Family	Source
Abui	abz	Trans-New Guinea	Kratochvil 2007
Achumawi	acv	Palaihnihan	de Angulo and Freeland 1931
Agta (Central)	agt	Austronesian	Healey 1960
Armenian (Western)	hyw	Indo-European	Gulian 1965
Avar	ava	Nakh-Daghestanian	Bokarev 1949b
Bontok	lbk	Austronesian	Waterman 1932
Didinga	did	Nilo-Saharan	Tucker and Bryan 1966
Diegueño (Mesa Grande)	dih	Cochimí-Yuman	Langdon 1970
Dinka	dip	Nilo-Saharan	Tucker and Bryan 1966
Enets	enf	Uralic	Künnap 1999a
Finnish	fin	Uralic	Sulkala and Karjalainen 1992
Futuna-Aniwa	fut	Austronesian	Dougherty 1983
Grebo	grj	Niger-Congo	Innes 1966
Halkomelem (Upriver)	hur	Salish	Galloway 1993
Hualapai	yuf	Cochimí-Yuman	Watahomigie et al. 1982
Ignaciano	ign	Maipurean	Ott and Ott 1967b
Katcha	xtc	Nilo-Saharan	Tucker and Bryan 1966
Koegu	xwg	Nilo-Saharan	Hieda 1998
Kolami	kfb	Dravidian	Emeneau 1955
Kombio	xbi	Torricelli	Yi 2008
Komi-Permyak	koi	Uralic	Riese 1998
Komi-Zyrian	kpv	Uralic	Hausenberg 1998
Komo	kmo	Niger-Congo	Tucker and Bryan 1966
Kyuquot	nuk	Wakashan	Rose 1981
Majang	mpe	Nilo-Saharan	Unseth 1989
Mano	mev	Niger-Congo	Becker-Donner 1965
Mari (Meadow)	mhr	Uralic	Sebeok and Ingemann 1961
Moro	mor	Niger-Congo	Black and Black 1971
Naga (Mao)	nbi	Sino-Tibetan	Giridhar 1994
Nahali	nlx	Indo-European	N/A
Nakanai	nak	Austronesian	Johnston 1980
Nenets	yrk	Uralic	Hajdú 1963, Salminen 1998

Nganasan	nio	Uralic	Helinski 1998a
Nisgha	ncg	Tsimshian	Tarpen 1987
Pokot	pko	Nilo-Saharan	Baroja 1989
Quileute	qui	Chimakuan	Andrade 1933
Rukai (Tanan)	dru	Austronesian	Li 1973
Saami (Northern)	sme	Uralic	Bartens 1989
Sanuma	xsu	Yanomaman	Borgman 1990
Selknam	ona	Chon	Najlis 1973
Siroi	ssd	Trans-New Guinea	Wells 1979
Squamish	squ	Salish	Kuipers 1967
Taiof	sps	Austronesian	Ross 2002g
Thai	tha	Kra-Dai	Anthony et al. 1968
Tsez	ddo	Nakh-Daghestanian	Alekseev and Radzhabov 2004
Udihe	ude	Tungusic	Nikolaeva and Tolskaya 2001
Zoque (Copainalá)	zoc	Mixe-Zoquean	Harrison et al. 1981

Table 4: Genetic affiliation and sources